

Mathematical Foundations of Loss Functions in Machine Learning

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Preface

This document develops the standard mathematical scaffolding around *loss functions* in machine learning. For each notion – predictors, loss, risk, empirical risk, Bayes optimality, regularisation – we (i) give a textbook definition, (ii) state the principal mathematical properties, and (iii) interpret the object in machine-learning practice.

1 Predictors, Losses, and Risk

A supervised-learning problem is specified by a data-generating distribution $\mathcal{P}$ on $\mathcal{X} \times \mathcal{Y}$, a predictor $f : \mathcal{X} \rightarrow \hat{\mathcal{Y}}$, and a *loss function*

$$L : \mathcal{Y} \times \hat{\mathcal{Y}} \longrightarrow \mathbb{R}_{\geq 0}.$$

The quantity to be minimised is the *risk*, i.e. the expected loss under the data distribution:

$$R(f) := \mathbb{E}_{(X,Y) \sim \mathcal{P}}[L(Y, f(X))] = \int_{\mathcal{X} \times \mathcal{Y}} L(y, f(x)) \, d\mathcal{P}(x, y).$$

Mathematical background.

Infimum and supremum.

Definition 1 (Infimum and supremum). For a nonempty set $S \subseteq \mathbb{R}$, the *supremum* $\sup S \in \mathbb{R} \cup \{+\infty\}$ is the least upper bound of S : (i) $s \leq \sup S$ for every $s \in S$, and (ii) for any $u < \sup S$ there exists $s \in S$ with $s > u$. The *infimum* is the greatest lower bound, defined dually as $\inf S := -\sup(-S)$. The completeness axiom of $\mathbb{R}$ guarantees that $\sup S$ and $\inf S$ exist (possibly $\pm\infty$). Unlike $\min$ and $\max$, an infimum or supremum need not itself belong to S .

Measurable spaces and measurable functions.

Definition 2 (σ -algebra, measurable space, measurable function). A σ -algebra on a set Ω is a collection $\mathcal{F} \subseteq 2^\Omega$ containing Ω and closed under complementation and countable unions. The pair $(\Omega, \mathcal{F})$ is a *measurable space*. A function $f : (\Omega_1, \mathcal{F}_1) \rightarrow (\Omega_2, \mathcal{F}_2)$ is *measurable* if $f^{-1}(A) \in \mathcal{F}_1$ for every $A \in \mathcal{F}_2$. For countable spaces we tacitly take $\mathcal{F} = 2^\Omega$ (all subsets are measurable), so measurability becomes a vacuous condition.

Probability spaces (Kolmogorov axioms).

Definition 3 (Probability space). A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ consisting of a sample space Ω , a σ -algebra $\mathcal{F}$ of events on Ω , and a *probability measure* $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ satisfying Kolmogorov's three axioms:

- (i) (*non-negativity*) $\mathbb{P}(A) \geq 0$ for every $A \in \mathcal{F}$;
- (ii) (*normalisation*) $\mathbb{P}(\Omega) = 1$;
- (iii) (*countable additivity*) for any pairwise disjoint family $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{F}$,

$$\mathbb{P}\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \sum_{i \in \mathbb{N}} \mathbb{P}(A_i).$$

Random variables and distributions.

Definition 4 (Random variable; pushforward distribution; joint and marginal). A *random variable* taking values in a measurable space $(\mathcal{X}, \mathcal{F}_{\mathcal{X}})$ is a measurable function $X : (\Omega, \mathcal{F}) \rightarrow (\mathcal{X}, \mathcal{F}_{\mathcal{X}})$. Its *distribution* (or *law*) is the *pushforward measure* $\mathcal{P}_X$ on $\mathcal{X}$ defined by

$$\mathcal{P}_X(A) := \mathbb{P}(X^{-1}(A)) = \mathbb{P}(X \in A), \quad A \in \mathcal{F}_{\mathcal{X}}.$$

For two random variables X, Y on the same probability space, their *joint distribution* $\mathcal{P}_{X,Y}$ is the pushforward of $\mathbb{P}$ by the map $\omega \mapsto (X(\omega), Y(\omega))$. The *marginal distribution* of X is recovered by integrating out Y : $\mathcal{P}_X(A) = \mathcal{P}_{X,Y}(A \times \mathcal{Y})$.

Independence.

Definition 5 (Independent random variables). Two random variables X, Y are *independent* if their joint distribution factorises into the product of marginals:

$$\mathcal{P}_{X,Y}(A \times B) = \mathcal{P}_X(A) \mathcal{P}_Y(B)$$

for all measurable $A \subseteq \mathcal{X}$ and $B \subseteq \mathcal{Y}$. Equivalently, for any pair of events $\{X \in A\}, \{Y \in B\}$, $\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A) \mathbb{P}(Y \in B)$. A family $X_1, \dots, X_n$ is *mutually independent* if the joint distribution factorises into the n -fold product of marginals.

Lebesgue integral.

Definition 6 (Lebesgue integral). The *Lebesgue integral* of a measurable function $f : (\Omega, \mathcal{F}) \rightarrow \mathbb{R}_{\geq 0}$ with respect to $\mathbb{P}$ is constructed in three stages.

- (1) (*Simple functions*) For $f = \sum_{i=1}^n a_i \mathbf{1}_{A_i}$ with $a_i \geq 0$ and disjoint $A_i \in \mathcal{F}$, $\int_{\Omega} f \, d\mathbb{P} := \sum_{i=1}^n a_i \mathbb{P}(A_i)$.

- (2) (*Non-negative functions*) For general non-negative measurable f ,

$$\int_{\Omega} f \, d\mathbb{P} := \sup \left\{ \int_{\Omega} s \, d\mathbb{P} : s \text{ simple, } 0 \leq s \leq f \right\}.$$

- (3) (*Signed functions*) For a general measurable f , write $f = f^+ - f^-$ with $f^{\pm} \geq 0$ and define $\int f \, d\mathbb{P} := \int f^+ \, d\mathbb{P} - \int f^- \, d\mathbb{P}$, provided not both pieces are infinite.

The Lebesgue integral coincides with the Riemann integral whenever both exist, but is defined for a vastly larger class of functions and satisfies strong convergence theorems (monotone, dominated, Fatou).

Expectation and conditional expectation.

Definition 7 (Expectation and conditional expectation). For a real-valued random variable Z on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, the *expectation* is the Lebesgue integral of Z against $\mathbb{P}$:

$$\mathbb{E}[Z] := \int_{\Omega} Z \, d\mathbb{P}.$$

On a countable set with PMF p , this reduces to $\mathbb{E}[Z] = \sum_z z p(z)$ when the series converges absolutely. The *conditional expectation* of Y given $X = x$ is

$$\mathbb{E}[Y \mid X = x] := \int y \, d\mathcal{P}_{Y|X=x}(y),$$

or $\sum_y y p(y \mid x)$ in the discrete case. The *law of total expectation* (tower property) states

$$\mathbb{E}[Y] = \mathbb{E}_X[\mathbb{E}[Y \mid X]].$$

Variance.

Definition 8 (Variance and standard deviation). For a real-valued random variable Z with $\mathbb{E}[Z^2] < \infty$ and mean $\mu := \mathbb{E}[Z]$, the *variance* is

$$\text{Var}(Z) := \mathbb{E}[(Z - \mu)^2] = \mathbb{E}[Z^2] - \mu^2 \geq 0.$$

It satisfies $\text{Var}(aZ + b) = a^2 \text{Var}(Z)$ and, for independent Z_1, Z_2 , $\text{Var}(Z_1 + Z_2) = \text{Var}(Z_1) + \text{Var}(Z_2)$. The *standard deviation* is $\sigma(Z) := \sqrt{\text{Var}(Z)}$.

Random data and predictors.

Definition 9 (Data-generating distribution and predictor). Let $(\mathcal{X}, \mathcal{F}_{\mathcal{X}})$ and $(\mathcal{Y}, \mathcal{F}_{\mathcal{Y}})$ be measurable spaces. The *data-generating distribution* is a probability measure $\mathcal{P}$ on the product space $\mathcal{X} \times \mathcal{Y}$; we write $(X, Y) \sim \mathcal{P}$. A *predictor* (or *hypothesis*) is a measurable function $f : \mathcal{X} \rightarrow \hat{\mathcal{Y}}$, where the *prediction space* $\hat{\mathcal{Y}}$ may differ from $\mathcal{Y}$ (e.g. probabilistic predictions for classification take values in the simplex $\Delta(\mathcal{Y})$). The collection of admissible predictors is the *hypothesis class* $\mathcal{H} \subseteq \hat{\mathcal{Y}}^{\mathcal{X}}$.

In Mathematics

A predictor is a measurable function, not just a function: measurability ensures that $f(X)$ is a random variable and that expectations of $L(Y, f(X))$ are well-defined. For finite or countable $\mathcal{X}, \mathcal{Y}$ this is automatic; for continuous spaces it is a non-trivial regularity assumption.

In Machine Learning

In supervised learning, the hypothesis class $\mathcal{H}$ is realised by a parameterised model family $\{f_{\theta} : \theta \in \Theta\}$ – linear models, decision trees, neural networks. Choosing $\mathcal{H}$ is the first design decision in any ML problem and trades off expressiveness against estimability.

Loss functions.

Definition 10 (Loss function). A *loss function* on target space $\mathcal{Y}$ and prediction space $\hat{\mathcal{Y}}$ is a measurable map

$$L : \mathcal{Y} \times \hat{\mathcal{Y}} \rightarrow \mathbb{R}_{\geq 0},$$

quantifying the cost incurred by predicting $\hat{y}$ when the true target is y . By convention $L(y, \hat{y}) = 0$ when the prediction exactly matches the target.

In Mathematics

The codomain is taken non-negative for two reasons: (i) it makes the risk $\mathbb{E}[L]$ well-defined as a Lebesgue integral, and (ii) it ensures that the optimisation problem $\min_f \mathbb{E}[L(Y, f(X))]$ has a well-defined infimum. Some authors allow signed losses, but then one must work with the integral of the positive and negative parts separately.

In Machine Learning

A loss is the operational specification of "what we are trying to do." A neural network with a different loss is a different algorithm – the same architecture trained with cross-entropy vs. squared loss behaves fundamentally differently. Choosing the loss is therefore the second design decision, immediately after $\mathcal{H}$.

Risk.

Definition 11 (Risk / generalisation error). The *risk* (or *expected loss*, or *generalisation error*) of a predictor f under the data-generating distribution $\mathcal{P}$ is

$$R(f) := \mathbb{E}_{(X,Y) \sim \mathcal{P}} [L(Y, f(X))] = \int_{\mathcal{X} \times \mathcal{Y}} L(y, f(x)) d\mathcal{P}(x, y).$$

In Mathematics

$R(\cdot)$ is a functional on the hypothesis class, $R : \mathcal{H} \rightarrow \mathbb{R}_{\geq 0}$. By the law of total expectation it factorises as

$$R(f) = \mathbb{E}_X [\mathbb{E}_{Y|X} [L(Y, f(X)) | X]],$$

which separates the dependence on the marginal $\mathcal{P}_X$ from the conditional $\mathcal{P}_{Y|X}$. This decomposition is the starting-point for the Bayes-optimal analysis in Section 6.

In Machine Learning

$R(f)$ is what we ultimately care about – the average loss on *future* data drawn from the same distribution. It is unobservable because $\mathcal{P}$ is unknown; everything in supervised learning is an attempt to estimate or upper-bound this quantity from finite data.

2 Empirical Risk and Empirical Risk Minimisation

Since $\mathcal{P}$ is unknown, $R(f)$ cannot be evaluated directly. Given an i.i.d. sample $\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^n$ from $\mathcal{P}$, the *empirical risk* is the sample-mean estimator of $R(f)$:

$$\hat{R}_n(f) := \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i)).$$

Empirical Risk Minimisation (ERM) selects the predictor minimising this empirical surrogate over a hypothesis class $\mathcal{H}$:

$$\hat{f}_n \in \arg \min_{f \in \mathcal{H}} \hat{R}_n(f).$$

The *generalisation gap* is the discrepancy $R(f) - \hat{R}_n(f)$ between true and empirical risk.

Mathematical background.

Independent and identically distributed samples.

Definition 12 (i.i.d. random variables). Random variables $Z_1, \dots, Z_n$ are *independent and identically distributed* (i.i.d.) under a distribution $\mathcal{P}$ if

- (i) each Z_i has marginal distribution $\mathcal{P}$;
- (ii) the joint distribution factorises: $\mathbb{P}(Z_1 \in A_1, \dots, Z_n \in A_n) = \prod_{i=1}^n \mathbb{P}(Z_i \in A_i)$ for all measurable $A_1, \dots, A_n$.

The joint distribution of $(Z_1, \dots, Z_n)$ is the *n-fold product measure* $\mathcal{P}^{\otimes n}$.

Empirical risk.

Definition 13 (Empirical risk). Given an i.i.d. training sample $\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^n \sim \mathcal{P}^{\otimes n}$, the *empirical risk* of a predictor f is

$$\hat{R}_n(f) := \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i)).$$

Proposition 14 (Unbiasedness and variance of $\hat{R}_n$). For any fixed f ,

$$\mathbb{E}[\hat{R}_n(f)] = R(f), \quad \text{Var}(\hat{R}_n(f)) = \frac{\text{Var}(L(Y, f(X)))}{n}.$$

Modes of stochastic convergence.

Definition 15 (Almost sure, in probability, in distribution). Let $Z_1, Z_2, \dots$ be real-valued random variables and Z another such. The sequence *converges to* Z :

- *almost surely*, written $Z_n \xrightarrow{\text{a.s.}} Z$, if $\mathbb{P}(\lim_{n \rightarrow \infty} Z_n = Z) = 1$;
- *in probability*, written $Z_n \xrightarrow{p} Z$, if for every $\varepsilon > 0$, $\mathbb{P}(|Z_n - Z| > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$;
- *in distribution*, written $Z_n \xrightarrow{d} Z$, if $\mathbb{E}[g(Z_n)] \rightarrow \mathbb{E}[g(Z)]$ for every bounded continuous function g .

The implications are $\xrightarrow{\text{a.s.}} \Rightarrow \xrightarrow{p} \Rightarrow \xrightarrow{d}$; the reverse implications fail in general.

Proposition 16 (Strong and weak laws of large numbers). Let $Z_1, Z_2, \dots$ be i.i.d. with $\mathbb{E}[|Z_1|] < \infty$ and mean $\mu = \mathbb{E}[Z_1]$. Then the sample mean $\bar{Z}_n := \frac{1}{n} \sum_{i=1}^n Z_i$ satisfies $\bar{Z}_n \xrightarrow{\text{a.s.}} \mu$ (*strong LLN*), which implies $\bar{Z}_n \xrightarrow{p} \mu$ (*weak LLN*).

Hoeffding's inequality.

Proposition 17 (Hoeffding's inequality). Let $Z_1, \dots, Z_n$ be independent random variables with $Z_i \in [a_i, b_i]$ almost surely, and let $\bar{Z}_n = \frac{1}{n} \sum_{i=1}^n Z_i$. Then for any $t > 0$,

$$\mathbb{P}\left(|\bar{Z}_n - \mathbb{E}[\bar{Z}_n]| > t\right) \leq 2 \exp\left(-\frac{2n^2 t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Applied to $Z_i = L(y_i, f(x_i))$ with all $Z_i \in [0, B]$, this gives the empirical-risk concentration bound

$$\mathbb{P}\left(|\hat{R}_n(f) - R(f)| > t\right) \leq 2 \exp\left(-\frac{2nt^2}{B^2}\right).$$

In Mathematics

$\hat{R}_n$ is the sample-mean estimator of R , so by the law of large numbers $\hat{R}_n(f) \rightarrow R(f)$ almost surely as $n \rightarrow \infty$. The $\mathcal{O}(1/\sqrt{n})$ concentration rate is the foundation of every finite-sample generalisation bound in statistical learning theory.

In Machine Learning

The empirical risk is what your training loop actually reports – it is the average per-example loss on the training set. Mini-batch SGD adds a second stochastic approximation: at each step, $\hat{R}_n$ is itself replaced by an average over a mini-batch, yielding an unbiased gradient estimate of $\hat{R}_n$.

The arg min and arg max operators.

Definition 18 (arg min and arg max). For a function $g : X \rightarrow \mathbb{R}$ on a nonempty set X , the *arg-minimum* and *arg-maximum* are the (possibly empty, possibly multi-valued) sets

$$\arg \min_{x \in X} g(x) := \{x_* \in X : g(x_*) \leq g(x) \text{ for all } x \in X\},$$

$$\arg \max_{x \in X} g(x) := \{x_* \in X : g(x_*) \geq g(x) \text{ for all } x \in X\}.$$

They are related by $\arg \min_x g(x) = \arg \max_x (-g(x))$, and both exist (are nonempty) whenever X is finite or whenever g is continuous and X is compact. When the set is a singleton, we abuse notation and identify arg min (resp. arg max) with its unique element.

Empirical risk minimisation and the generalisation gap.

Definition 19 (ERM). The *empirical risk minimiser* over a hypothesis class $\mathcal{H}$ is

$$\hat{f}_n \in \arg \min_{f \in \mathcal{H}} \hat{R}_n(f).$$

The *generalisation gap* of a predictor f is $R(f) - \hat{R}_n(f)$.

In Mathematics

Bounding the generalisation gap uniformly over $\mathcal{H}$ – that is, controlling $\sup_{f \in \mathcal{H}} |R(f) - \hat{R}_n(f)|$ – is the central problem of statistical learning theory. Capacity measures such as VC dimension, Rademacher complexity, and PAC-Bayes bounds all serve this purpose.

In Machine Learning

Most training procedures (gradient descent on a loss) are approximations of ERM. Overfitting is the empirical phenomenon by which $\hat{R}_n(\hat{f}_n)$ becomes small while $R(\hat{f}_n)$ remains large, i.e. the generalisation gap is large. This motivates regularisation (§7).

3 Regression Losses

For real-valued targets $\mathcal{Y} = \hat{\mathcal{Y}} = \mathbb{R}$, the three canonical losses are

$$L_{\text{sq}}(y, \hat{y}) = (y - \hat{y})^2 \quad (\text{squared loss}),$$

$$L_{\text{abs}}(y, \hat{y}) = |y - \hat{y}| \quad (\text{absolute loss}),$$

$$L_{\text{Huber}}^\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2, & |y - \hat{y}| \leq \delta, \\ \delta |y - \hat{y}| - \frac{1}{2}\delta^2, & |y - \hat{y}| > \delta, \end{cases} \quad (\text{Huber loss with threshold } \delta > 0).$$

All three are convex; the first is C^∞ , the second is non-smooth at zero, and the third is the unique C^1 interpolation between them.

Mathematical background.

Convex and strictly convex functions.

Definition 20 (Convex, strictly convex, smooth). A function $g : C \rightarrow \mathbb{R}$ on a convex set $C \subseteq \mathbb{R}^d$ is *convex* if for all $x, x' \in C$ and $\lambda \in [0, 1]$

$$g(\lambda x + (1 - \lambda)x') \leq \lambda g(x) + (1 - \lambda)g(x').$$

It is *strictly convex* if the inequality is strict whenever $x \neq x'$ and $\lambda \in (0, 1)$, and is *L-smooth* (with constant $L > 0$) if ∇g is L -Lipschitz: $\|\nabla g(x) - \nabla g(x')\| \leq L\|x - x'\|$. For twice-differentiable g on a convex domain, convexity is equivalent to $\nabla^2 g(x) \succeq 0$. If $\nabla^2 g(x) \succ 0$ everywhere, then g is strictly convex, but the converse need not hold.

In Mathematics

Convexity is the structural property under which any local minimum of g is also a global minimum, and the gradient ∇g has the variational characterisation $g(x') \geq g(x) + \langle \nabla g(x), x' - x \rangle$. These two facts underlie essentially all of convex optimisation.

In Machine Learning

A loss function inherits convexity from being convex in its prediction argument $\hat{y}$. Convex losses give well-behaved optimisation – no spurious local minima, monotone progress under gradient descent – but real-world ML losses (e.g. deep-network training losses) are typically non-convex in the parameters θ even when the underlying loss $L(y, \cdot)$ is convex.

Smoothness class C^k .

Definition 21 (C^k class). A function $g : U \rightarrow \mathbb{R}$ on an open set $U \subseteq \mathbb{R}^d$ belongs to the class C^k , for $k \in \{0, 1, 2, \dots, \infty\}$, if it has continuous partial derivatives up to order k . In particular, C^0 means continuous, C^1 means continuously differentiable, C^2 means twice continuously differentiable, and C^∞ means smooth (infinitely differentiable). For a single real variable, $g \in C^k$ iff $g, g', \dots, g^{(k)}$ all exist and are continuous.

Squared loss.

Definition 22 (Squared loss).

$$L_{\text{sq}}(y, \hat{y}) := (y - \hat{y})^2.$$

In Mathematics

L_{sq} is the squared distance under the Euclidean metric on $\mathbb{R}$. As a function of $\hat{y}$ it is strictly convex, infinitely differentiable, and has constant second derivative 2; the corresponding risk $R(f) = \mathbb{E}[(Y - f(X))^2]$ is a quadratic functional. By Pythagoras of $L^2(\mathcal{P})$, the Bayes-optimal predictor is the conditional mean $f^*(x) = \mathbb{E}[Y \mid X = x]$ (see §6).

In Machine Learning

Squared loss is the default for regression: it yields closed-form solutions in linear models (ordinary least squares), Gaussian likelihood interpretation, and tractable optimisation in neural nets (MSE loss). Its main weakness is high sensitivity to outliers, since each error is squared.

Absolute loss.

Definition 23 (Absolute loss).

$$L_{\text{abs}}(y, \hat{y}) := |y - \hat{y}|.$$

In Mathematics

L_{abs} is convex but not differentiable at $\hat{y} = y$. The Bayes-optimal predictor for absolute loss is the conditional median $\text{Median}(Y \mid X = x)$. The corresponding empirical procedure is least-absolute-deviations (LAD) regression, which is more robust to outliers than OLS but requires linear-programming solvers.

In Machine Learning

Absolute loss is used when residuals are heavy-tailed or when one wants robustness to outliers. Quantile loss $L_{\tau}(y, \hat{y}) := \tau(y - \hat{y})_+ + (1 - \tau)(\hat{y} - y)_+$ generalises it: the Bayes-optimal predictor becomes the τ -th conditional quantile, enabling quantile regression.

Huber loss.

Definition 24 (Huber loss). For a threshold $\delta > 0$,

$$L_{\text{Huber}}^{\delta}(y, \hat{y}) := \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{if } |y - \hat{y}| \leq \delta, \\ \delta |y - \hat{y}| - \frac{1}{2}\delta^2 & \text{if } |y - \hat{y}| > \delta. \end{cases}$$

In Mathematics

Huber loss is the convex combination – in fact the unique C^1 function – that is quadratic for small residuals and linear for large ones. It is C^1 but not C^2 ; its derivative is bounded by δ , providing gradient clipping for free.

In Machine Learning

Huber loss interpolates between squared loss (efficient near the optimum) and absolute loss (robust to outliers). It is the standard choice in robust regression and is used in DQN-style reinforcement learning to stabilise Bellman-error targets.

4 Classification Losses

For discrete targets, the two canonical losses are

$$L_{0-1}(y, \hat{y}) = \mathbf{1}\{\hat{y} \neq y\} = \begin{cases} 0, & \hat{y} = y, \\ 1, & \hat{y} \neq y, \end{cases} \quad (0-1 \text{ loss}),$$

$$L_{\text{hinge}}(y, \hat{y}) = \max\{0, 1 - y\hat{y}\} \quad (\text{hinge loss, for } \mathcal{Y} = \{-1, +1\}, \hat{y} \in \mathbb{R}).$$

L_{0-1} is the quantity we ultimately wish to minimise (the misclassification rate), but is non-convex and non-differentiable; L_{hinge} is a convex *surrogate* that upper-bounds it under sign decoding.

Mathematical background.

Indicator function.

Definition 25 (Indicator function). For a set X and a predicate $P(x)$ on X (equivalently, a subset $A \subseteq X$), the *indicator function* is

$$\mathbf{1}\{P(x)\} = \mathbf{1}_A(x) := \begin{cases} 1, & \text{if } P(x) \text{ is true (equivalently, } x \in A), \\ 0, & \text{otherwise.} \end{cases}$$

It takes values in $\{0, 1\}$, and its expectation under a distribution p is the probability of the predicate: $\mathbb{E}_{x \sim p}[\mathbf{1}\{P(x)\}] = \mathbb{P}(P(X))$.

Subgradient and subdifferential.

Definition 26 (Subdifferential). For a convex function $g : \mathbb{R}^d \rightarrow \mathbb{R}$ and a point x_0 , the *subdifferential* of g at x_0 is the set

$$\partial g(x_0) := \{v \in \mathbb{R}^d : g(x) \geq g(x_0) + \langle v, x - x_0 \rangle \text{ for all } x \in \mathbb{R}^d\}.$$

Elements of $\partial g(x_0)$ are called *subgradients*. If g is differentiable at x_0 , then $\partial g(x_0) = \{\nabla g(x_0)\}$; at non-smooth points (e.g. the kink of $|x|$ at $x = 0$), the subdifferential is a non-singleton interval. A point x_0 is a global minimiser of g iff $0 \in \partial g(x_0)$.

Zero-one loss.

Definition 27 (0-1 loss).

$$L_{0-1}(y, \hat{y}) := \mathbf{1}\{\hat{y} \neq y\} = \begin{cases} 0 & \hat{y} = y, \\ 1 & \hat{y} \neq y. \end{cases}$$

In Mathematics

L_{0-1} is non-convex, non-continuous, and almost-everywhere flat; its gradient is zero wherever defined. Direct minimisation of empirical 0-1 risk is NP-hard for most non-trivial hypothesis classes. The Bayes-optimal predictor is the conditional mode $f^*(x) = \arg \max_y \mathcal{P}(Y = y \mid X = x)$.

In Machine Learning

L_{0-1} is what we ultimately want to minimise for classification – it is exactly the misclassification rate – but the gradient-zero property makes it unusable in gradient-based training. This motivates *surrogate losses*: convex upper bounds on L_{0-1} that are tractable to optimise.

Hinge loss.

Definition 28 (Hinge loss). For $\mathcal{Y} = \{-1, +1\}$ and real-valued predictions $\hat{y} \in \mathbb{R}$,

$$L_{\text{hinge}}(y, \hat{y}) := \max\{0, 1 - y\hat{y}\}.$$

In Mathematics

Hinge loss is convex, piecewise linear, and a surrogate (in fact, an upper bound) for L_{0-1} under the sign-decoding rule $\hat{y}_{\text{class}} = \text{sign}(\hat{y})$. Its subdifferential at the kink $y\hat{y} = 1$ contains 0, which gives rise to the support-vector phenomenon: only points near the margin contribute non-zero subgradients.

In Machine Learning

Hinge loss is the defining loss of the support-vector machine (SVM). The "margin" $y\hat{y} \geq 1$ inside which the loss is zero is the geometric origin of the SVM's max-margin principle. Multi-class extensions (Crammer–Singer hinge) generalise this to $\mathcal{Y} = \{1, \dots, K\}$.

5 Probabilistic Losses: NLL, Cross-Entropy, KL

When the prediction is a probability distribution $\hat{p} \in \Delta(\mathcal{Y})$, three information-theoretic losses arise:

$$L_{\text{NLL}}(y, \hat{p}) = -\log \hat{p}(y) \quad (\text{negative log-likelihood}),$$

$$H(p, \hat{p}) = -\sum_{y \in \mathcal{Y}} p(y) \log \hat{p}(y) \quad (\text{cross-entropy}),$$

$$\text{KL}(p \parallel \hat{p}) = \sum_{y \in \mathcal{Y}} p(y) \log \frac{p(y)}{\hat{p}(y)} \quad (\text{KL divergence}).$$

They are linked by the identity $H(p, \hat{p}) = H(p) + \text{KL}(p \parallel \hat{p})$, where $H(p) = -\sum p \log p$ is the entropy. On a one-hot target $p = \delta_{y^*}$, all three reduce to $-\log \hat{p}(y^*)$.

Mathematical background.

Probability simplex.

Definition 29 (Probability simplex). For a countable set $\mathcal{Y}$, the *probability simplex* over $\mathcal{Y}$ is the set of all probability mass functions on it:

$$\Delta(\mathcal{Y}) := \left\{ p : \mathcal{Y} \rightarrow [0, 1] \mid \sum_{y \in \mathcal{Y}} p(y) = 1 \right\}.$$

For $|\mathcal{Y}| = K$ finite, $\Delta(\mathcal{Y})$ is the $(K-1)$ -dimensional standard simplex embedded in $\mathbb{R}^K$: $\Delta_{K-1} = \{p \in \mathbb{R}^K : p_k \geq 0, \sum_k p_k = 1\}$.

Dirac point mass.

Definition 30 (Dirac point mass). For an element $y^* \in \mathcal{Y}$, the *Dirac point mass* (or *one-hot distribution*) at y^* is the PMF

$$\delta_{y^*}(y) := \mathbf{1}\{y = y^*\} = \begin{cases} 1, & y = y^*, \\ 0, & y \neq y^*. \end{cases}$$

It is the extreme point of $\Delta(\mathcal{Y})$ supported at y^* . On a probability space, the corresponding measure is $\delta_{y^*}(A) := \mathbf{1}\{y^* \in A\}$.

Gibbs' inequality and Cramér–Rao lower bound.

Proposition 31 (Gibbs' inequality). For two PMFs $p, \hat{p}$ on a countable set $\mathcal{Y}$ with $\hat{p}(y) > 0$ wherever $p(y) > 0$,

$$\text{KL}(p \parallel \hat{p}) \geq 0,$$

with equality if and only if $p = \hat{p}$. Equivalently, $H(p, \hat{p}) \geq H(p)$ – the cross-entropy is bounded below by the entropy of p .

Definition 32 (Fisher information matrix). For a parametric family $\{p_\theta : \theta \in \Theta \subseteq \mathbb{R}^d\}$ satisfying standard regularity conditions (twice differentiability in θ , support not depending on θ , interchangeability of differentiation and integration), the *Fisher information matrix* at θ is

$$I(\theta) := \mathbb{E}_{Y \sim p_\theta} \left[(\nabla_\theta \log p_\theta(Y)) (\nabla_\theta \log p_\theta(Y))^\top \right] = -\mathbb{E}_{Y \sim p_\theta} [\nabla_\theta^2 \log p_\theta(Y)].$$

Proposition 33 (Cramér–Rao lower bound). Under the regularity conditions above, any unbiased estimator $\hat{\theta}_n$ of θ based on n i.i.d. observations satisfies

$$\text{Cov}(\hat{\theta}_n) \succeq \frac{1}{n} I(\theta)^{-1},$$

where $\succeq$ denotes the Löwner (positive-semidefinite) order. An unbiased estimator attaining this bound is called *efficient*; the maximum-likelihood estimator is asymptotically efficient under the same conditions.

Negative log-likelihood.

Definition 34 (NLL). For a target $y \in \mathcal{Y}$ and a predicted distribution $\hat{p} \in \Delta(\mathcal{Y})$,

$$L_{\text{NLL}}(y, \hat{p}) := -\log \hat{p}(y).$$

In Mathematics

Minimising the empirical NLL over a parametric family $\{\hat{p}_\theta\}$ is the principle of *maximum likelihood estimation* (MLE): $\arg \min_\theta \sum_i -\log \hat{p}_\theta(y_i | x_i) = \arg \max_\theta \sum_i \log \hat{p}_\theta(y_i | x_i)$. Under regularity conditions, the MLE is consistent, asymptotically normal, and asymptotically efficient (attains the Cramér–Rao lower bound).

In Machine Learning

NLL is the most-used loss in modern ML. Language-model pre-training, classification heads, generative models (VAEs, diffusion, autoregressive models) all optimise variants of NLL. It is the natural loss whenever the model outputs a probability distribution.

Cross-entropy.

Definition 35 (Cross-entropy). For two distributions $p, \hat{p}$ on the same countable set $\mathcal{Y}$,

$$H(p, \hat{p}) := -\sum_{y \in \mathcal{Y}} p(y) \log \hat{p}(y).$$

Proposition 36 (Cross-entropy reduces to NLL on one-hot targets). If $p = \delta_{y^*}$ is the point mass at y^* , then $H(\delta_{y^*}, \hat{p}) = -\log \hat{p}(y^*) = L_{\text{NLL}}(y^*, \hat{p})$.

In Mathematics

$H(p, \hat{p}) = H(p) + \text{KL}(p \parallel \hat{p})$, where $H(p) := -\sum p \log p$ is the entropy of p . Since $H(p)$ does not depend on $\hat{p}$, minimising cross-entropy in $\hat{p}$ is equivalent to minimising the forward KL divergence $\text{KL}(p \parallel \hat{p})$.

In Machine Learning

In classification, the empirical distribution of labels is one-hot, so “cross-entropy loss” and “NLL” are used interchangeably in practice. The PyTorch `CrossEntropyLoss` module is, technically, the composition of softmax + NLL with one-hot targets.

Kullback–Leibler divergence as a loss.

Definition 37 (KL divergence).

$$\text{KL}(p \parallel \hat{p}) := \sum_y p(y) \log \frac{p(y)}{\hat{p}(y)} = \mathbb{E}_{y \sim p} \left[\log \frac{p(y)}{\hat{p}(y)} \right].$$

In Mathematics

$\text{KL}(p \parallel \hat{p}) \geq 0$ with equality iff $p = \hat{p}$ (Gibbs' inequality). It is not symmetric: $\text{KL}(p \parallel \hat{p}) \neq \text{KL}(\hat{p} \parallel p)$ in general, and the two have qualitatively different behaviours – “forward” KL is mass-covering, “reverse” KL is mode-seeking. The Pythagorean identity $H(p, \hat{p}) = H(p) + \text{KL}(p \parallel \hat{p})$ links cross-entropy and KL.

In Machine Learning

KL appears as a loss in variational inference (ELBO contains $\text{KL}(q \parallel p)$), as a regulariser in RLHF and DPO (KL-regularised reward maximisation), and as a distillation loss ($\text{KL}(p_{\text{teacher}} \parallel p_{\text{student}})$). The asymmetry matters: choosing forward vs. reverse KL changes both the optimisation landscape and the qualitative shape of the learned distribution.

6 Bayes Risk and Bayes-Optimal Predictors

The *Bayes risk* is the smallest achievable risk over all measurable predictors:

$$R^* := \inf_{f \text{ measurable}} R(f).$$

A predictor attaining this infimum is called *Bayes-optimal*. For each canonical loss, the Bayes-optimal predictor is a known functional of the conditional distribution $\mathcal{P}_{Y|X}$:

$$L_{\text{sq}} \implies f^*(x) = \mathbb{E}[Y \mid X = x] \quad (\text{conditional mean}),$$

$$L_{\text{abs}} \implies f^*(x) = \text{Median}(Y \mid X = x) \quad (\text{conditional median}),$$

$$L_{0-1} \implies f^*(x) = \arg \max_{y \in \mathcal{Y}} \mathcal{P}(Y = y \mid X = x) \quad (\text{conditional mode}),$$

$$L_{\text{NLL}} \implies \hat{p}^*(\cdot \mid x) = \mathcal{P}(\cdot \mid X = x) \quad (\text{true conditional distribution}).$$

Mathematical background.

Definition 38 (Conditional median and conditional mode). For a real-valued random variable Y with conditional distribution $\mathcal{P}_{Y|X=x}$, a *conditional median* is any value $m \in \mathbb{R}$ satisfying

$$\mathbb{P}(Y \leq m \mid X = x) \geq \frac{1}{2} \quad \text{and} \quad \mathbb{P}(Y \geq m \mid X = x) \geq \frac{1}{2}.$$

For discrete $\mathcal{Y}$, the *conditional mode* is any maximiser of the conditional PMF:

$$\text{Mode}(Y \mid X = x) \in \arg \max_{y \in \mathcal{Y}} \mathbb{P}(Y = y \mid X = x).$$

Neither the conditional median nor the conditional mode is required to be unique.

Definition 39 (Bayes risk and Bayes-optimal predictor). The *Bayes risk* is the infimum of the risk over all measurable predictors:

$$R^* := \inf_{f \text{ measurable}} R(f).$$

A *Bayes-optimal predictor* is any f^* attaining the infimum, $R(f^*) = R^*$.

Proposition 40 (Bayes-optimal predictors for canonical losses). Under mild regularity,

- (i) *Squared loss*: $f^*(x) = \mathbb{E}[Y \mid X = x]$ (conditional mean).

- (ii) *Absolute loss*: $f^*(x) = \text{Median}(Y \mid X = x)$ (conditional median).
- (iii) *0-1 loss*: $f^*(x) = \arg \max_{y \in \mathcal{Y}} \mathcal{P}(Y = y \mid X = x)$ (conditional mode).
- (iv) *Cross-entropy / NLL*: $\hat{p}^*(\cdot \mid x) = \mathcal{P}(\cdot \mid X = x)$ (the true conditional distribution).

In Mathematics

Each Bayes-optimal predictor can be derived by minimising the conditional risk $\mathbb{E}[L(Y, \hat{y}) \mid X = x]$ pointwise in $\hat{y}$. For squared loss this is a quadratic in $\hat{y}$ minimised at the conditional mean; for cross-entropy it is a strictly convex functional on the simplex minimised at the true conditional distribution by Gibbs' inequality.

In Machine Learning

Bayes optimality is the irreducible benchmark: no predictor can do better on the population, no matter how powerful the hypothesis class. The *excess risk* $R(f) - R^*$ decomposes into approximation error ($\inf_{\mathcal{H}} R - R^*$) and estimation error ($R(\hat{f}_n) - \inf_{\mathcal{H}} R$). Larger $\mathcal{H}$ reduces the first and increases the second – the bias–variance trade-off.

7 Regularised Risk and Structural Risk Minimisation

Adding a complexity penalty $\Omega(f) \geq 0$ to the empirical risk yields the *regularised empirical risk*:

$$\hat{R}_{n,\lambda}(f) := \hat{R}_n(f) + \lambda \Omega(f), \quad \lambda \geq 0.$$

For parametric models f_θ with $\theta \in \mathbb{R}^d$, the canonical penalties are the ℓ^p norms,

$$\Omega_p(\theta) = \|\theta\|_p^p = \sum_{j=1}^d |\theta_j|^p,$$

with $p = 2$ (*ridge*, Tikhonov regularisation, weight decay) and $p = 1$ (*lasso*, sparsity-inducing) by far the most widely used.

Mathematical background.

Regularised empirical risk.

Definition 41 (Regularised empirical risk). For a complexity penalty $\Omega : \mathcal{H} \rightarrow \mathbb{R}_{\geq 0}$ and regularisation strength $\lambda \geq 0$,

$$\hat{R}_{n,\lambda}(f) := \hat{R}_n(f) + \lambda \Omega(f).$$

The *structural risk minimiser* is $\hat{f}_{n,\lambda} \in \arg \min_{f \in \mathcal{H}} \hat{R}_{n,\lambda}(f)$.

ℓ^p norms.

Definition 42 (ℓ^p norm). For $\theta \in \mathbb{R}^d$ and $p \in [1, \infty)$, the ℓ^p norm is

$$\|\theta\|_p := \left(\sum_{j=1}^d |\theta_j|^p \right)^{1/p}.$$

The limiting cases are $\|\theta\|_\infty := \max_j |\theta_j|$ and, for $p < 1$, $\|\cdot\|_p$ is a quasi-norm. For $p \geq 1$, $\|\cdot\|_p$ satisfies the three norm axioms (positivity, absolute homogeneity, triangle inequality), is convex, and is strictly convex iff $p > 1$.

Probability density functions.

Definition 43 (Probability density function). For a probability measure μ on $\mathbb{R}^d$, a *probability density function* (PDF) with respect to the Lebesgue measure $d\theta$ is a non-negative measurable function $p : \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}$ such that

$$\mu(A) = \int_A p(\theta) d\theta \quad \text{for every measurable } A \subseteq \mathbb{R}^d.$$

By the Radon–Nikodym theorem, such a density exists if and only if μ is *absolutely continuous* with respect to the Lebesgue measure – i.e. $\mu(A) = 0$ whenever the Lebesgue measure of A is zero. The total mass satisfies $\int_{\mathbb{R}^d} p(\theta) d\theta = 1$.

Maximum a posteriori estimation.

Definition 44 (MAP estimate). Given a prior PDF π on a parameter $\theta \in \mathbb{R}^d$ and a likelihood $p_\theta(\mathcal{D})$ of observed data $\mathcal{D}$, Bayes’ rule gives the *posterior*

$$\pi(\theta \mid \mathcal{D}) = \frac{p_\theta(\mathcal{D}) \pi(\theta)}{\int p_{\theta'}(\mathcal{D}) \pi(\theta') d\theta'} \propto p_\theta(\mathcal{D}) \pi(\theta).$$

The *maximum-a-posteriori* (MAP) estimate is the posterior mode:

$$\hat{\theta}_{\text{MAP}} := \arg \max_{\theta \in \mathbb{R}^d} \pi(\theta \mid \mathcal{D}) = \arg \max_{\theta} \left[\log p_\theta(\mathcal{D}) + \log \pi(\theta) \right].$$

When the prior is uniform, $\hat{\theta}_{\text{MAP}}$ reduces to the maximum-likelihood estimate.

Gaussian and Laplace distributions.

Definition 45 (Multivariate Gaussian). The *multivariate Gaussian distribution* $\mathcal{N}(\mu, \Sigma)$ on $\mathbb{R}^d$, with mean $\mu \in \mathbb{R}^d$ and positive-definite covariance $\Sigma \in \mathbb{R}^{d \times d}$, has density

$$\mathcal{N}(\theta; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\theta - \mu)^\top \Sigma^{-1}(\theta - \mu)\right).$$

For isotropic covariance $\Sigma = \sigma^2 I$, the negative log-density equals $\frac{1}{2\sigma^2} \|\theta - \mu\|_2^2$ up to an additive constant.

Definition 46 (Laplace distribution). The *Laplace distribution* with location $\mu \in \mathbb{R}$ and scale $b > 0$ on $\mathbb{R}$ has density

$$\text{Laplace}(\theta; \mu, b) = \frac{1}{2b} \exp\left(-\frac{|\theta - \mu|}{b}\right).$$

For an i.i.d. Laplace prior on $\theta \in \mathbb{R}^d$ with $\mu = 0$, the negative log-density equals $\frac{1}{b} \|\theta\|_1$ up to an additive constant.

Common penalties (Tikhonov / ridge, lasso).

Definition 47 (ℓ^p penalties). For a parametric model f_θ with $\theta \in \mathbb{R}^d$, the ℓ^p penalty is the p -th power of the ℓ^p norm:

$$\Omega_p(\theta) := \|\theta\|_p^p = \sum_{j=1}^d |\theta_j|^p.$$

The cases $p = 2$ (*ridge*, Tikhonov) and $p = 1$ (*lasso*) are canonical.

Proposition 48 (Bayesian interpretation). For a model $\hat{p}_\theta(y \mid x)$ trained by MLE with NLL, adding a penalty $\lambda \Omega(\theta)$ corresponds to a maximum a posteriori (MAP) estimate under a prior $\pi(\theta) \propto \exp(-\lambda \Omega(\theta))$:

- $\Omega(\theta) = \frac{1}{2} \|\theta\|_2^2$ corresponds to a Gaussian prior $\mathcal{N}(0, \lambda^{-1} I)$.

- $\Omega(\theta) = \|\theta\|_1$ corresponds to an i.i.d. Laplace prior with scale $1/\lambda$.

In Mathematics

ℓ_2 penalty preserves convexity and differentiability; the regularised quadratic problem has closed form $\hat{\theta} = (X^\top X + \lambda I)^{-1} X^\top y$ or $\hat{\theta} = (X^\top X + n\lambda I)^{-1} X^\top y$, depending on the scaling convention. Under the empirical-risk convention $\frac{1}{n} \sum_i (y_i - x_i^\top \theta)^2 + \lambda \|\theta\|_2^2$, the latter is the consistent form. ℓ_1 is convex but non-differentiable at the origin; its subdifferential includes 0 in a coordinate direction whenever $|\partial \hat{R}_n / \partial \theta_j| \leq \lambda$, which produces exact sparsity, i.e. zero coefficients.

In Machine Learning

ℓ^2 regularisation is called *weight decay* in deep learning and is equivalent (for SGD with fixed learning rate) to multiplicatively shrinking weights each step. ℓ^1 regularisation is used for sparse feature selection. Other "structural" regularisers (dropout, data augmentation, early stopping) admit similar Bayesian or implicit interpretations and are studied as instances of structural risk minimisation.

Summary. The supervised learning problem has four moving parts:

- a *data distribution* $\mathcal{P}$ on $\mathcal{X} \times \mathcal{Y}$;
- a *hypothesis class* $\mathcal{H}$ of measurable predictors;
- a *loss function* $L : \mathcal{Y} \times \hat{\mathcal{Y}} \rightarrow \mathbb{R}_{\geq 0}$;
- a *learning rule* that selects some $\hat{f} \in \mathcal{H}$ from data.

The loss defines the *risk* $R(f) = \mathbb{E}_{\mathcal{P}}[L(Y, f(X))]$, which the learning rule approximates via the *empirical risk* $\hat{R}_n(f) = \frac{1}{n} \sum_i L(y_i, f(x_i))$, optionally penalised by $\lambda \Omega(f)$. The *Bayes risk* R^* is the distribution-dependent lower bound on R ; for canonical losses, the Bayes-optimal predictor has a closed form – conditional mean for squared loss, conditional median for absolute, conditional mode for 0-1, and the true conditional distribution for cross-entropy.

Every named loss in this document is a different choice within the same framework. Choosing a loss is, mathematically, choosing which functional of the conditional distribution $\mathcal{P}_{Y|X}$ the Bayes-optimal predictor recovers; choosing a regulariser is, equivalently, choosing a prior over the hypothesis class. Every other ML loss (contrastive, preference-based DPO, RL-style policy-gradient surrogates) can be read as an instance or extension of this same scaffold.

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